

Two Experiments, One Quantity

Linear independence, matrix equations and the Jacobian

Reading route and mathematical level

CORE ROUTE

The physical argument uses equations, substitution and proportional reasoning. It can be read without knowing matrices.

MATRIX EXTENSION

The 2×2 matrix, determinant and rank-nullity language form an optional extension. These ideas should not be assumed as standard MYP knowledge, nor as universal prior knowledge across DP mathematics courses.

UNIVERSITY PREVIEW

The Jacobian uses partial derivatives of a function of two variables. It is a first-year-university preview, included to show how the same question is asked for a general nonlinear system.

The question

On the surface of an unfamiliar planet, we perform two experiments:

- drop an object through a known height h and measure its fall time t ;
- use a pendulum of known length ℓ and measure its period τ .

The planet has unknown mass M and unknown radius R . Two measured times give two equations. Is that enough to determine the two unknowns separately?

We assume that the drop height and pendulum length are small compared with the planetary radius, so that the local field is approximately uniform. We also use the small-angle approximation for the pendulum.

The two gravitational clocks

CORE ROUTE

This section is the main physical route.

At the planet's surface,

$$g = \frac{GM}{R^2}, \quad (1)$$

where G is the gravitational constant.

For an object released from rest,

$$h = \frac{1}{2}gt^2,$$

and hence

$$t^2 = \frac{2h}{g}. \quad (2)$$

For a small-angle pendulum,

$$\tau = 2\pi\sqrt{\frac{\ell}{g}},$$

so

$$\tau^2 = \frac{4\pi^2\ell}{g}. \quad (3)$$

The notation carries a physical distinction. The symbol t describes the duration of a one-way event; τ describes the duration of one repeating cycle. Nevertheless, both times contain the same local quantity g .

Indeed, the two experiments separately give

$$g_{\text{fall}} = \frac{2h}{t^2}, \quad g_{\text{pend}} = \frac{4\pi^2\ell}{\tau^2}. \quad (4)$$

Agreement between these values supports the model. It does not yet follow that the measurements contain two independent pieces of information about the planet.

Core conclusion without matrices

Both experiments determine g . Since $g = GM/R^2$, they therefore determine

$$\frac{M}{R^2} = \frac{g}{G},$$

rather than M and R separately. The second apparatus can corroborate the first measurement, but it does not automatically create a second independent piece of information.

Worked numerical example: an Earth-like surface

Take

$$g = 9.81 \text{ m s}^{-2}, \quad h = 1.00 \text{ m}, \quad \ell = 1.00 \text{ m}.$$

The measured times predicted by the two models are

$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2.00}{9.81}} \approx 0.4515 \text{ s},$$

$$\tau = 2\pi\sqrt{\frac{\ell}{g}} = 2\pi\sqrt{\frac{1.00}{9.81}} \approx 2.006 \text{ s}.$$

Putting either result back into its equation recovers the same field strength:

$$\frac{2h}{t^2} \approx 9.81 \text{ m s}^{-2}, \quad \frac{4\pi^2\ell}{\tau^2} \approx 9.81 \text{ m s}^{-2}.$$

The two numbers look quite different because one is a one-way fall and the other is a complete oscillation. Their planetary information is nevertheless the same.

From simultaneous equations to a matrix equation**MATRIX EXTENSION**

The remainder of this section assumes elementary 2×2 matrix notation.

Before returning to the planet, consider a general pair of linear equations:

$$\begin{aligned} aX + bY &= p, \\ cX + dY &= q. \end{aligned}$$

They can be collected into one matrix equation:

$$\underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{\text{coefficient matrix}} \underbrace{\begin{pmatrix} X \\ Y \end{pmatrix}}_{\text{unknown vector}} = \underbrace{\begin{pmatrix} p \\ q \end{pmatrix}}_{\text{output vector}}. \quad (5)$$

Each row of the coefficient matrix represents one equation. The equations are independent when neither row can be made by multiplying the other by a single number.

Geometrically, the absolute value of a 2×2 determinant is the factor by which the matrix scales area. A zero determinant means that a two-dimensional region has been flattened onto a line or a point: one direction has been lost. Algebraically, that same collapse means that the two rows are dependent. For a 2×2 matrix,

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Determinant test

If $ad - bc \neq 0$, the coefficient matrix is invertible and the two equations determine one unique pair (X, Y) . If $ad - bc = 0$, the rows are linearly dependent, so the equations cannot determine a unique pair.

Rewriting the planetary equations

Substitute $g = GM/R^2$ into the falling-object equation:

$$\begin{aligned} t^2 &= \frac{2h}{GM/R^2} \\ &= \frac{2hR^2}{GM}. \end{aligned}$$

Multiplying by GM and bringing all terms to the left gives

$$(Gt^2)M - (2h)R^2 = 0. \quad (6)$$

The same steps for the pendulum give

$$\begin{aligned} \tau^2 &= \frac{4\pi^2\ell}{GM/R^2} \\ &= \frac{4\pi^2\ell R^2}{GM}, \end{aligned}$$

and therefore

$$(G\tau^2)M - (4\pi^2\ell)R^2 = 0. \quad (7)$$

The original equations are nonlinear in M and R . They become linear if we temporarily use the transformed unknowns

$$X = M, \quad Y = R^2.$$

This substitution works because, after clearing the denominator, R appears only through the single power R^2 . It is a special feature of this power-law relationship, not a recipe that turns every nonlinear system into a linear one. The Jacobian extension later in the note gives the general method for making a *local* linear approximation to a nonlinear map.

Equations (6) and (7) can then be written as

$$\underbrace{\begin{pmatrix} Gt^2 & -2h \\ G\tau^2 & -4\pi^2\ell \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} M \\ R^2 \end{pmatrix}}_{\mathbf{x}} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (8)$$

This is a *homogeneous* matrix equation, $\mathbf{A}\mathbf{x} = \mathbf{0}$.

Are the two rows independent?

The determinant of the coefficient matrix is

$$\det(\mathbf{A}) = (Gt^2)(-4\pi^2\ell) - (-2h)(G\tau^2) \quad (9)$$

$$= G(2h\tau^2 - 4\pi^2\ell t^2). \quad (10)$$

Both experiments occur in the same gravitational field, so the two estimates in (4) must agree:

$$\frac{2h}{t^2} = \frac{4\pi^2\ell}{\tau^2}.$$

Cross-multiplication gives

$$2h\tau^2 = 4\pi^2\ell t^2. \quad (11)$$

Substitution into (10) now gives

$$\boxed{\det(\mathbf{A}) = 0.}$$

The dependence can also be seen directly. Equation (11) implies

$$\frac{\tau^2}{t^2} = \frac{2\pi^2\ell}{h} = \lambda,$$

and therefore

$$\begin{pmatrix} G\tau^2 & -4\pi^2\ell \end{pmatrix} = \lambda \begin{pmatrix} Gt^2 & -2h \end{pmatrix}.$$

The second row is a multiple of the first. The matrix has rank 1, rather than rank 2.

Numerical check of the determinant

For the worked example $h = \ell = 1.00$ m, $t = 0.4515$ s and $\tau = 2.006$ s. Since $G \neq 0$, it is enough to check the bracket in Equation (10):

$$\begin{aligned}\frac{\det(\mathbf{A})}{G} &= 2h\tau^2 - 4\pi^2\ell t^2 \\ &\approx 2(1.00)(2.006)^2 - 4\pi^2(1.00)(0.4515)^2 \\ &\approx 8.04807 - 8.04776 \\ &\approx 0.\end{aligned}$$

The small residual comes from rounding t and τ . With the unrounded model values, the determinant is exactly zero.

Homogeneous systems and rank-nullity

The right-hand side of (8) is zero, so this is a homogeneous system. For a matrix with two columns,

$$\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = 2.$$

Here the *rank* counts independent rows and the *nullity* counts independent directions satisfying $\mathbf{A}\mathbf{x} = \mathbf{0}$.

If $\det(\mathbf{A}) \neq 0$, then the rank is 2 and the nullity is 0, so the only solution is $\mathbf{x} = \mathbf{0}$. That would mean $M = 0$ and $R^2 = 0$. Here $\det(\mathbf{A}) = 0$ and $\mathbf{A} \neq \mathbf{0}$, so the rank is 1 and the nullity is 1. The physically allowed solutions form a one-parameter family.

What have the experiments actually determined?

Either row of the matrix equation gives

$$\frac{M}{R^2} = \frac{2h}{Gt^2} = \frac{4\pi^2\ell}{G\tau^2} = \frac{g}{G}. \quad (12)$$

Thus the observations determine the combination M/R^2 , rather than M and R separately. Every point on the curve

$$M = \frac{g}{G}R^2 \quad (13)$$

describes a planet producing the same values of t and τ .

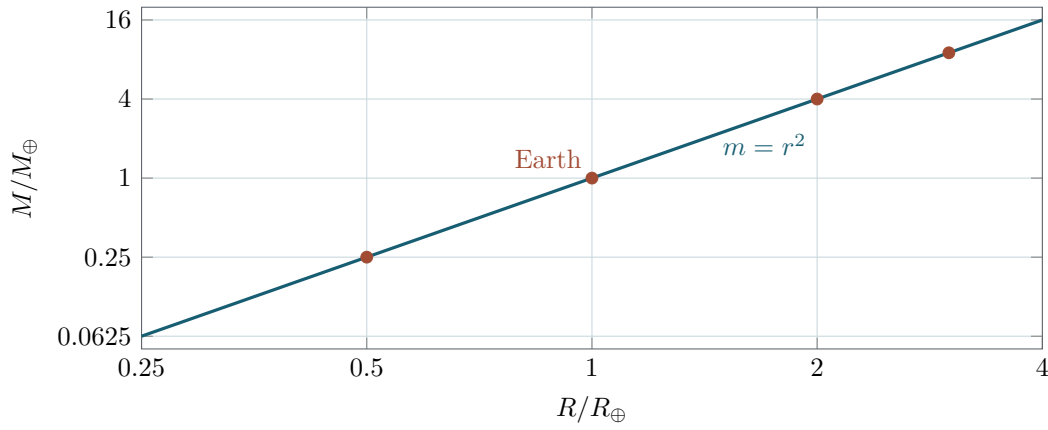
For example, if R is multiplied by a factor c , then M can be multiplied by c^2 without changing either experiment:

$$R \mapsto cR, \quad M \mapsto c^2M, \quad \frac{GM}{R^2} \mapsto \frac{G(c^2M)}{(cR)^2} = \frac{GM}{R^2}.$$

For an Earth-strength field, introduce the dimensionless variables

$$r = \frac{R}{R_\oplus}, \quad m = \frac{M}{M_\oplus}.$$

The degeneracy curve becomes simply $m = r^2$. On logarithmic axes it is a straight line of gradient 2:



Every point on the line represents a different pair (M, R) with the same surface value of g . A change of one decade horizontally corresponds to two decades vertically.

The lesson in linear independence

Counting equations is not the same as counting information. Different apparatus can generate different observations while probing the same combination of unknown quantities. The pendulum corroborates the falling-object measurement of g , but it does not separate M from R .

In a real experiment, measurement uncertainty would usually make g_{fall} and g_{pend} slightly different. Combining them could improve an estimate of g or expose weaknesses in the models. It would still not identify M and R separately: repeated evidence for one quantity does not become a second independent quantity.

University preview: the Jacobian

UNIVERSITY PREVIEW This section uses partial derivatives and is included as a conceptual bridge beyond school-level calculus.

The determinant above tests linear independence after introducing the transformed variable R^2 . The Jacobian provides the corresponding test for the original nonlinear map from planetary parameters to measured times.

Define

$$a = \sqrt{\frac{2h}{G}}, \quad b = 2\pi\sqrt{\frac{\ell}{G}}.$$

Then the two predicted measurements are

$$t(M, R) = aRM^{-1/2}, \quad \tau(M, R) = bRM^{-1/2}. \quad (14)$$

The Jacobian collects the partial derivatives of the observations with respect to the unknown parameters:

$$\mathbf{J} = \frac{\partial(t, \tau)}{\partial(M, R)} = \begin{pmatrix} \frac{\partial t}{\partial M} & \frac{\partial t}{\partial R} \\ \frac{\partial \tau}{\partial M} & \frac{\partial \tau}{\partial R} \end{pmatrix}. \quad (15)$$

From (14),

$$\frac{\partial t}{\partial M} = -\frac{t}{2M}, \quad \frac{\partial t}{\partial R} = \frac{t}{R},$$

and similarly

$$\frac{\partial \tau}{\partial M} = -\frac{\tau}{2M}, \quad \frac{\partial \tau}{\partial R} = \frac{\tau}{R}.$$

Consequently,

$$\mathbf{J} = \begin{pmatrix} -\frac{t}{2M} & \frac{t}{R} \\ -\frac{\tau}{2M} & \frac{\tau}{R} \end{pmatrix}. \quad (16)$$

Its determinant is

$$\begin{aligned}\det(\mathbf{J}) &= \left(-\frac{t}{2M}\right) \left(\frac{\tau}{R}\right) - \left(\frac{t}{R}\right) \left(-\frac{\tau}{2M}\right) \\ &= -\frac{t\tau}{2MR} + \frac{t\tau}{2MR} \\ &= 0.\end{aligned}$$

For small changes, the Jacobian acts like a local coefficient matrix:

$$\begin{pmatrix} \Delta t \\ \Delta \tau \end{pmatrix} \approx \mathbf{J} \begin{pmatrix} \Delta M \\ \Delta R \end{pmatrix}. \quad (17)$$

If $\det(\mathbf{J}) \neq 0$, independent small changes in M and R could produce distinguishable changes in the measurements. Here $\det(\mathbf{J}) = 0$: there is a direction in parameter space that the two experiments cannot see.

Along the family $M \propto R^2$,

$$\Delta M \approx \frac{2M}{R} \Delta R.$$

Indeed,

$$\mathbf{J} \begin{pmatrix} 2M/R \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Moving in this direction changes M and R while leaving both t and τ unchanged to first order. In this example the degeneracy is exact, not merely local.

The logarithmic view

Logarithms make the shared dependence especially visible. Let

$$u = \ln M, \quad v = \ln R.$$

Equation (14) becomes

$$\begin{pmatrix} \ln(t/a) \\ \ln(\tau/b) \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & 1 \\ -\frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}. \quad (18)$$

The two rows are identical. Both logarithmic measurements contain only the combination

$$v - \frac{1}{2}u = \ln\left(\frac{R}{\sqrt{M}}\right).$$

The ordinary determinant and the Jacobian determinant are therefore revealing the same structural fact at two different levels.

How could the degeneracy be broken?

We need a measurement with a genuinely different dependence on M and R . For example, measure gravity at the surface and again at a known height H :

$$g_0 = \frac{GM}{R^2}, \quad g_H = \frac{GM}{(R+H)^2}.$$

Their ratio contains R :

$$\sqrt{\frac{g_0}{g_H}} = \frac{R+H}{R} = 1 + \frac{H}{R}.$$

Hence

$$R = \frac{H}{\sqrt{g_0/g_H} - 1}, \quad M = \frac{g_0 R^2}{G}.$$

This additional experiment probes how the field changes with distance. It adds a new dependence and raises the rank from one to two.

Your turn: does a third clock help?

For each proposed experiment, ask a structural question before doing any calculation:

Does its equation contain a new combination of M and R ?

1. **CORE ROUTE** Set $h = \ell$. Use the two time equations to show that

$$\frac{\tau}{t} = \sqrt{2} \pi.$$

Estimate this ratio and identify the quantities that cancel.

2. **CORE ROUTE** Planet B has twice the radius of planet A. The fall and pendulum experiments give the same results on both planets. How must their masses be related?
3. **CORE ROUTE** An ideal mass–spring clock has period

$$T_s = 2\pi \sqrt{\frac{m_s}{k}},$$

where m_s is the attached mass and k is the spring constant. Would its period break the M – R degeneracy? Explain from the equation.

4. **MATRIX EXTENSION** A vertical spring also has an equilibrium extension $x_0 = m_s g/k$. If m_s and k are known, what planetary combination does measuring x_0 determine? Does the rank rise to 2?
5. **MATRIX EXTENSION** A Foucault pendulum has an ideal precession period

$$T_F = \frac{2\pi}{\Omega |\sin \phi|},$$

where Ω is the planet's rotation rate and ϕ is latitude. Does this clock separate M and R without an additional model connecting rotation to gravitation?

6. **UNIVERSITY PREVIEW** On an ideal airless planet, a satellite skimming just above the surface has orbital period

$$T_o = 2\pi \sqrt{\frac{R^3}{GM}}.$$

Combine this with $g = GM/R^2$ to obtain separate expressions for R and M . Why is this clock genuinely independent of the two surface clocks?

Answers for self-check

1. $\tau/t = \sqrt{2}\pi \approx 4.44$. The common length, g , and therefore the hidden combination GM/R^2 , cancel.
2. $R_B = 2R_A$ requires $M_B = 4M_A$, because M/R^2 must remain constant.
3. No. The ideal spring period contains neither M nor R . A new-looking clock is not automatically a new planetary measurement.
4. Since $g = kx_0/m_s$, the static extension again determines only $M/R^2 = g/G$. Its row is dependent on the existing gravitational information, so the rank remains 1.
5. No. It measures the additional parameter Ω (if ϕ is known), but supplies no new relation between M and R . More measured quantities do not help if they concern different unknowns.
6. Substitute $GM = gR^2$ into the orbital equation:

$$T_o^2 = \frac{4\pi^2 R^3}{GM} = \frac{4\pi^2 R}{g}.$$

Therefore

$$R = \frac{gT_o^2}{4\pi^2}, \quad M = \frac{gR^2}{G}.$$

The orbital period contains R^3/M , whereas the surface clocks contain R^2/M . The different powers provide an independent constraint.

Final principle

Two equations can determine two unknowns only when the equations are independent. The determinant tests this for a linear system. The Jacobian tests it locally for a nonlinear system.